blades. The blades cover the stator and rotor of the end plates.



Fig. 4.4 **Blade**

iv. Bearings

The bearings are placed on the top and bottom of the fan, used to reduce friction in rotating parts and also to reduce the noise while in motion.



Fig. 4.5 **Bearing**

v. Down rod

The down rod is made up of hard steel, according to the length required.

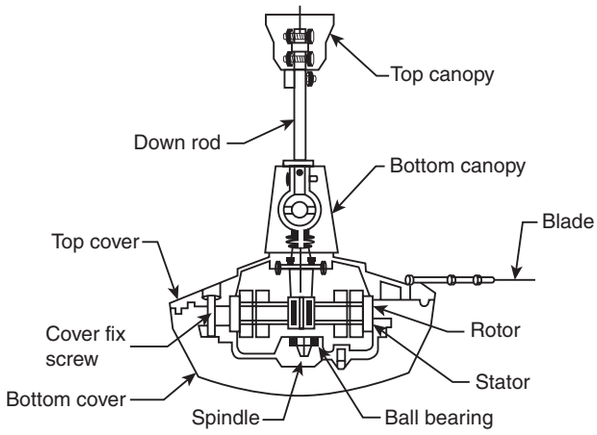


Fig. 4.6 **Internal structure of the fan**

One side of rod is fitted in ceiling hook, and the other side is fitted to the fan.

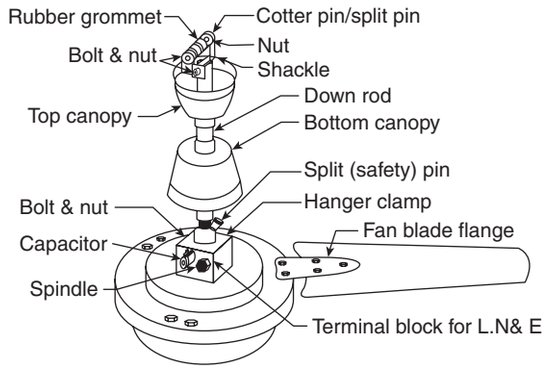


Fig. 4.7 **External structure of the fan**

a) Working principle

Figure 4.8 shows the internal structure of an electric fan. When an electric supply is given to the fan, the current passes to the main winding and auxiliary winding and produces rotating magnetic field. The 2.5 micro farad capacitor is connected in series with the auxiliary winding. Due to the production of rotating magnetic field, the blades connected with the rotor rotate and air flow will be circulated to the area where required. Usually the blades are available in various sizes like 900 mm, 1050 mm, 1200 mm, and 1400 mm respectively.

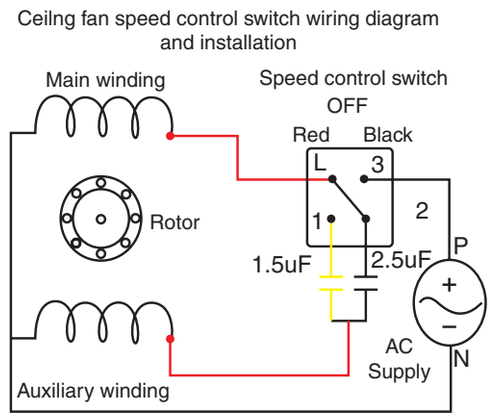


Fig. 4.8 **Circuit diagram**

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